

Electrophilic addition reaction of alkenes (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R3.4.11) Describe and explain the mechanisms of the reactions between symmetrical alkenes and halogens, water and hydrogen halides.
- (R3.4.12) Predict and explain the major product of a reaction between an unsymmetrical alkene and a hydrogen halide or water.

One of the applications of electrophilic addition of alkenes is producing ethanol from ethene. The process involves the reaction of ethene with concentrated sulfuric acid to produce ethyl hydrogensulfate, which then is diluted with water. Ethanol then is produced through a distillation process. The sulfuric acid here is used as a catalyst and is reproduced in the process.

The analogy of electrophilic addition reactions can be displayed by imagining two parents, C_1 and C_2 have lost sight of their children, X_1 and X_2 at a busy park (**Interactive 1**). The parents are anxiously looking for their children and hold both hands tightly with each other. Meanwhile, the children are wandering around the park and also holding both hands with each other. As soon as they spot their children, the parents rush towards them with relief. However, the parents face a dilemma as they do not want to let go of each other or lose track of their children again. The eventual solution is that C_1 lets go of one of their hands and takes the of X_1 , while X_2 quickly grabs the hand of C_2 . In this sequence, the family is finally reunited. Think of it as a reaction of an electrophile with alkenes, where the parents are carbon-carbon double bonds and the children are halogen atoms. The reunion is the result of a sequence that breaks the pi bond and forms the halogenoalkane.

In **Interactive 1**, click through the images to view the stages.

Interactive 1. Electrophilic Addition Reactions Analogy.

 More information for interactive 1

An interactive with five sections with title “Electron addition reactions analogy”.

Section 1. An illustration of a park scene with multiple people engaging in various activities. In the foreground, two individuals labeled C₁ and C₂ are holding both hands on a path. To the right, several people are jogging, and two individuals are playing football. In the middle ground, a woman is walking while looking at her phone. Further back, a person in a wheelchair is accompanied by another person standing nearby. On the right side, two individuals labeled X₁ and X₂ are walking together, while a man is jogging past them. In the background, two people are lying on a mat. The park contains multiple trees and bushes, with paths branching in different directions.

Section 2. In the given park scene, the two individuals labeled X₁ and X₂ walk towards the two individuals holding hands labeled C₁ and C₂.

Section 3. Now the two individuals labeled X₁ and X₂ walk together but do not hold hands anymore as they walk towards the two individuals holding only one hand labeled C₁ and C₂.

Section 4. The four labelled individuals are walking in a line, holding hands. The formation from left to right is as follows. X₁, C₁, C₂, X₂.

This analogy is applied in reactions of alkenes that are either symmetrical or unsymmetrical in general, which is similar to that described in section R3.4.4–5 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrophilic-addition-reaction-id-46329/>) where it focuses more on how the addition reaction will reduce the degree of unsaturated compound and form a new compound by involving an electrophile. In this section, the additional reactions of alkenes focus more on the alkene structures (symmetrical and

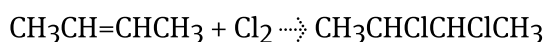
unsymmetrical), as these produce specific products. What are the differences between symmetrical and unsymmetrical alkenes? What about their products? Are they different?

Symmetrical alkenes

When alkenes undergo an electrophilic addition reaction, either end of the original alkenes are being added to by the new substituents. The products when the alkenes are symmetrical and when they are unsymmetrical are not identical. This is because each end of the double bond in symmetrical alkenes have the same substituents whereas unsymmetrical alkenes do not.

Reaction with halides

The addition of halogens, water or hydrogen halides to symmetrical alkenes produces only one type of product. For example, the reaction between but-2-ene and chlorine produces 2,3-dichlorobutane is shown in **Interactive 2**, next.



The mechanism starts when the alkene repels the electrons in the chlorine molecule, because if the double bond is an electron-rich a chlorine molecule is polarised. The chlorine molecule then is split heterolytically and forms a chloride ion causing the chlorine atom near the carbon atom to become electron deficient (an electrophile).

As a result, two π electrons move to form σ bond between carbon with the nearer chlorine atom, producing a carbocation. Therefore, chloride ions act as nucleophiles and donate a pair of electrons to the carbocation intermediate and form a coordination bond.

Click through the stages of **Interactive 2** to view the reaction mechanism.

Interactive 2. Electrophilic Addition of Chlorine to but-2-ene.

 More information for interactive 2

The interactive slideshow illustrates the electrophilic addition of chlorine to but-2-ene in six different slides. In all six slides, there is a text at the top that states “Electrophilic addition of chlorine to but-2-ene.”

Read below to learn what happens in each slide:

Slide 1:

The central structure shows but-2-ene, which consists of four carbon atoms. The double bond is located between the second and third carbon atoms. On the left of the double bond is a methyl group (CH_3 , first carbon) attached to the second carbon. On the right is another methyl group (CH_3 , fourth carbon) attached to the third carbon. Each of the two central carbon atoms in the double bond also has one hydrogen atom attached.

A chlorine molecule ($\text{Cl}-\text{Cl}$) is positioned below the double bond of the but-2-ene molecule. The chlorine atom closer to the double bond of 2-butene is labelled δ^+ (delta plus) in light blue. The second chlorine atom, farther from the double bond, is labelled δ^- (delta minus) in pink and has a pair of electrons toward it. This chlorine molecule is labelled as an “induced dipole”.

Slide 2:

This slide shows the interaction step before the reaction proceeds. The central structure shows but-2-ene and is the same as explained in slide 1. The chlorine molecule is also located below the double bond and looks similar as shown in slide 1. However, there aren't any electrons shown near the chlorine molecule in this slide. A curved arrow from the double bond points toward the δ^+ chlorine atom.

Slide 3:

Slide 3 shows the but-2-ene and Cl_2 molecule and they look similar to the ones in slide 2. The only difference is that there is a second arrow. The second arrow points from the $\text{Cl}-\text{Cl}$ bond toward the δ^- chlorine and there is a caption on the right side of Cl_2 that reads “Split heterolytically”.

Slide 4:

This image represents the intermediate stage in the electrophilic addition of chlorine (Cl_2) to but-2-ene, showing what happens after the double bond breaks and the molecule begins to rearrange. The main structure in the image is a modified form of but-2-ene, a four-carbon molecule. The double bond between the second and third carbon atoms has been broken. The second carbon is now bonded to a hydrogen atom (H), the first carbon which is a methyl group (CH_3) and a chlorine atom (Cl) shown in blue. The third carbon is bonded to a hydrogen atom (H) and fourth carbon atom which is a methyl group (CH_3). Additionally, the third carbon now carries a positive charge ($^+$), indicating that it has become a carbocation — a carbon atom that has lost an electron. A chloride ion (Cl^-) is shown nearby, in pink with a pair of dots (representing lone pair electrons), indicating that it is negatively charged and no longer bonded to any other atom. A note to the right of the image reads: “The pi bond breaks”.

Slide 5:

This slide focuses on the interaction between a carbocation and a chloride ion (Cl^-). The slide shows the modified form of but-2-ene, the same as shown in the previous slide. However, a chloride ion (Cl^-) is shown in pink with a lone pair of electrons (two pink dots), and it is positioned close to the positively charged carbon. A curved arrow points from the lone pair of electrons on Cl^- toward the positively charged carbon, indicating that the chloride ion is donating a pair of electrons to form a new bond, stabilizing the carbocation. Slide 6:

This image shows the full reaction mechanism for the electrophilic addition of chlorine (Cl_2) to but-2-ene, presented in three distinct structures (steps) from top to bottom. In the first structure (top left), the but-2-ene molecule is shown with a Cl_2 molecule beneath the double bond. There are two curved arrow marks: one between the double bond and the δ^+ chlorine atom, and the other from the $\text{Cl}-\text{Cl}$ bond toward the δ^- chlorine.

There is an arrow mark towards the right of this structure pointing towards the next step on the top right. In the second step, the structure shown is exactly similar to the one explained in slide 5. Beneath this structure, there is an arrow pointing downwards.

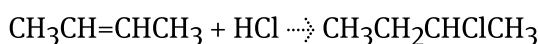
Below this arrow, there is a third structure indicating the product formation. This is the final molecule in which both formerly double-bonded carbon atoms now have a chlorine atom attached: One in blue (originally from Cl_2), one in pink (originally the Cl^- ion). This forms 2, 3-dichlorobutane, the final stable product.

This reaction mechanism in the slideshow involves the three main steps of electrophilic addition: Polarization and attack on Cl_2 , carbocation formation and chloride ion approach, and final bonding to form a stable saturated product.

Through this interactive viewers gain insight into heterolytic bond cleavage, carbocation formation, and nucleophilic attack, reinforcing core concepts in organic reaction mechanisms.

Reaction with hydrogen halides

The mechanism reaction between an alkene and hydrogen halide is similar to the previous mechanism. For example, the reaction between but-2-ene and hydrogen chloride produces 2-chlorobutane as shown in **Interactive 3**.



In this reaction, but-2-ene contains a double bond, which has a strong σ bond and a weak π bond, that acts as a nucleophile donating two pi electrons to an electrophile. Which atom would be attacked first by the pi electrons in but-2-ene?

Hydrogen chloride has a polar hydrogen—chloride bond in which the hydrogen atom is an electrophilic centre. Therefore, the pi electrons from the carbon double bond in but-2-ene attack the hydrogen atom of

hydrogen chloride, leading to the formation of a new bond.

The attraction of a pair of electrons in the but-2-ene by the hydrogen chloride as an electrophile causes a pair of electrons in the covalent bond between the hydrogen and chloride atom to move onto the chloride ion. The hydrogen–chloride bond splits heterolytically as the hydrogen atom can only form one bond and forms chloride ions. As the π bond no longer exists, this allows the formation of a carbocation intermediate and chloride ion. The electron pair on the negatively charged chloride ion is attracted to the positive carbon atom in the carbocation leading to the addition product.

Click through the images to view the stages of the reaction mechanism in **Interactive 3**.

Interactive 3. Electrophilic Addition of Hydrogen Chloride to but-2-ene.

 More information for interactive 3

This interactive guides users through the stepwise mechanism of the electrophilic addition of hydrogen chloride (HCl) to but-2-ene. The process is broken into six sections, which users can explore one at a time using navigation buttons given below.

Section 1: Chemical structure of but-2-ene. The double bond between the two central C atoms to. Chemical structure of HCl. The hydrogen has a delta positive charge and is bonded to the Cl group with a delta negative sign and a pair of electrons towards it. It is labeled as dipole-dipole.

Section 2. Chemical structure of but-2-ene. An arrow points from the double bond between the two central C atoms to an H atom with a delta positive sign. The hydrogen is bonded to the Cl group with a delta negative sign.

Section 3. Chemical structure of but-2-ene. An arrow points from the double bond between the two central C atoms to an H atom with a delta positive sign. The hydrogen is bonded to the Cl group with a delta negative sign. An

arrow points from the bond to the Cl group and is labeled Split heterolytically. Section 4. Chemical structure. C1 single bond C2. C1 is bonded to two H atoms and a CH₃ atom. One H atom is highlighted. C2 has a positive charge and is single-bonded to H and CH₃ groups.

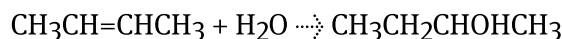
Section 5. Chemical structure. C1 single bond C2. C1 is bonded to two H atoms and a CH₃ atom. One H atom is highlighted. C2 has a positive charge and is single-bonded to H and CH₃ groups. A Cl group with a negative charge and a lone pair moves to the C2 atom.

Section 5. The overall chemical reaction. The reaction follows Markovnikov's rule, where the hydrogen (H) from HCl attaches to the carbon with more existing hydrogen atoms, forming a carbocation intermediate. The chloride group (Cl⁻) then attaches to the positively charged carbon. The final product formed is C1 single bond C2. C1 is bonded to two H atoms and a CH₃ group. C2 is bonded to a H atom, a Cl atom, and a CH₃ group.

This interactive effectively visualizes each mechanistic step, reinforcing concepts of polar bond interactions, carbocation stability, and regioselectivity in electrophilic additions.

Reaction with water

There is a similar mechanism in the reaction between alkene and water. For example, the reaction between but-2-ene and water producing butan-2-ol is shown in **Interactive 4**.



In this reaction, but-2-ene as a nucleophile attacks hydrogen atoms that have a polar hydrogen–oxygen bond. Therefore, two π electrons from the carbon double bond are used, leading to the formation of a new bond with the hydrogen atom causing the hydrogen atom bond to split heterolytically. This results in the formation of a carbocation intermediate and hydroxide ions. The hydroxide ion as a nucleophile then reacts quickly with the carbocation intermediate producing the alcohol.

Click through the stages of **Interactive 4** to view the reaction mechanism.

Interactive 4. Electrophilic Addition of Water to but-2-ene.

 More information for interactive 4

The interactive illustrates the electrophilic addition of water to but-2-ene across five sequentially numbered sections, indicated by a slide navigation bar at the bottom of the screen.

Section 1. Chemical structure of but-2-ene. An arrow points from the double bond between the two central C atoms to an H atom with a delta positive sign. The hydrogen is bonded to the OH group with a delta negative sign.

Section 2. Chemical structure of but-2-ene. An arrow points from the double bond between the two central C atoms to an H atom with a delta positive sign. The hydrogen is bonded to the OH group with a delta negative sign. An arrow points from the bond to the OH group and is labeled Split heterolytically.

Section 3. Chemical structure. C₁ single bond C₂. C₁ is bonded to two H atoms and a CH₃ atom. One H atom is highlighted. C₂ has a positive charge and is single-bonded to H and CH₃ groups.

Section 4. Chemical structure. C₁ single bond C₂. C₁ is bonded to two H atoms and a CH₃ atom. One H atom is highlighted. C₂ has a positive charge and is single-bonded to H and CH₃ groups. An OH group with a negative charge and a lone pair moves to the C₃ atom.

Section 5. The overall chemical reaction. The reaction follows Markovnikov's rule, where the hydrogen (H) from water attaches to the carbon with more existing hydrogen atoms, forming a carbocation intermediate. The hydroxyl group (OH⁻) then attaches to the positively charged carbon, resulting in the formation of butan-2-ol as the final product.

The reaction mechanism involves the initial polarization of the double bond, leading to protonation, carbocation formation, and nucleophilic attack by the hydroxide ion.

A more detailed mechanism is shown in **Figure 1**. This mechanism shows the H^+ ion acting as a catalyst, which is known as the acid-catalysed mechanism.

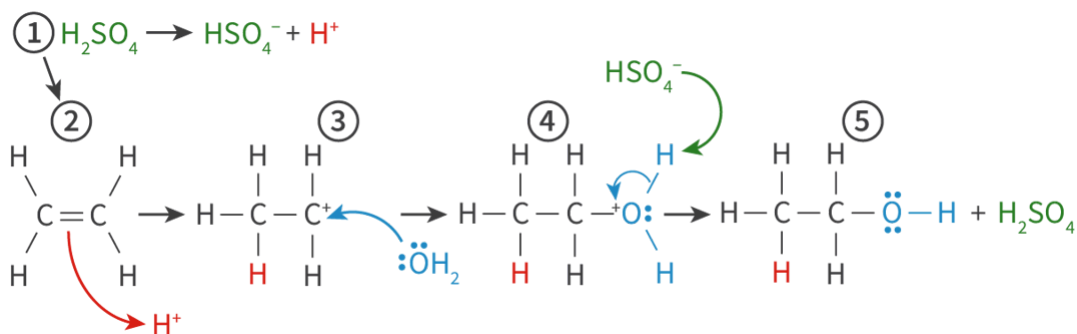
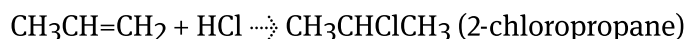


Figure 1. The acid-catalysed mechanism.

1. Sulfuric acid, H_2SO_4 , is the catalyst in the reaction and provides hydrogen ions (H^+) that act as the electrophile.
2. The H^+ ion bonds to one of the carbon atoms of the carbon-carbon double bond, forming a carbocation intermediate.
3. A water molecule bonds with the carbocation intermediate, forming a protonated ethanol molecule.
4. The HSO_4^- ion removes a proton from the protonated ethanol molecule, which reforms the sulfuric acid catalyst.
5. The product of the reaction is ethanol, $\text{C}_2\text{H}_5\text{OH}$ (in addition to the reformed sulfuric acid catalyst).

Unsymmetrical alkene reactions

On the other hand, two different products are produced when unsymmetrical alkenes react with a hydrogen halide: for example, the reaction between propene with hydrogen chloride. In this reaction, two different products are formed depending on where the halide and hydrogen atoms are attached.



The stability of carbocation determines the favoured product, which helps to predict the major product of the electrophilic addition. The more alkyl groups attached to the carbon centre, the more stable the carbocation (**Figure 2**). In [section R3.4.9–10](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-mechanism-hl-id-46331/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-mechanism-hl-id-46331/>), a tertiary carbon is shown to be more stable due to more alkyl groups attached to the carbon centre.

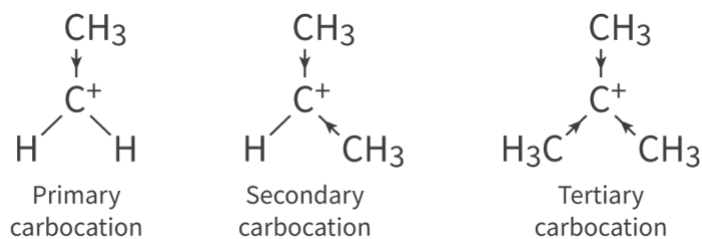


Figure 2. Stability of a carbocation.

 More information for figure 2

The image presents three structural formulas depicting a primary, secondary, and tertiary carbocation. Each structure includes a central carbon atom marked with a positive charge.

To the left, the primary carbocation is shown with the carbon atom bonded to a single methyl group (CH₃) and two hydrogen atoms (H).

In the center, the secondary carbocation features the carbon bonded to one methyl group (CH₃), one hydrogen atom (H), and another carbon bonded to a methyl group (CH₃), creating more stability than the primary.

On the right, the tertiary carbocation shows the carbon atom bonded to three separate methyl groups (CH₃), creating the most stable carbocation among the three displayed due to the highest number of alkyl groups attached.

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Study skills

Charged species are more stable when the charge can be spread out over more than one atom. A positive charge can be better dispersed over a C—C bond than a C—H bond. This is because the alkyl groups are larger and contain more electrons than an H atom, allowing their own electron density to be drawn toward the positive carbon, effectively spreading out the charge. Therefore, tertiary carbocation intermediates are more effective at spreading out the positive charge than a primary carbocation intermediate.

When propene reacts with hydrogen chloride, theoretically there will be two different products formed that are isomers of each other (**Figure 3**). This leads to two possibilities of pathways through electrophilic addition reactions.

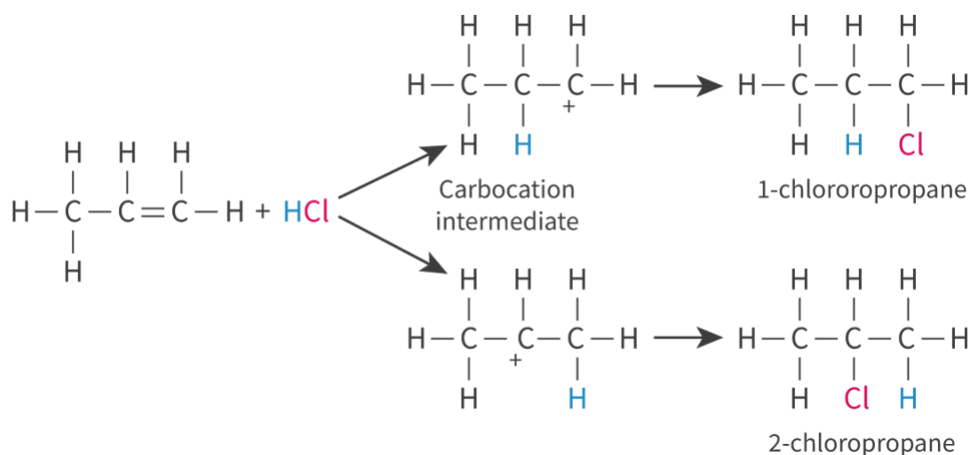


Figure 3. An electrophilic addition reaction of hydrogen chloride to propene.

 More information for figure 3

The diagram illustrates the electrophilic addition reaction of hydrogen chloride (HCl) to propene (C₃H₆). Starting at the left, the structure of propene is depicted as H₂C=CH-CH₃. Hydrogen chloride (HCl) is added to this molecule. Two possible carbocation intermediates are formed.

The first pathway splits upward, where propene reacts by adding H⁺ to the terminal carbon, creating a carbocation on the central carbon. From here, Cl⁻ attacks the positively charged center, forming 1-chloropropane (CH₃-CH₂-CH₂Cl).

The second pathway splits downward, where H⁺ adds to the central carbon, creating a carbocation on the terminal carbon. Here, Cl⁻ attacks this positively charged carbon, leading to the formation of 2-chloropropane (CH₃-CHCl-CH₃).

Both pathways are shown with arrows, indicating possible products, 1-chloropropane and 2-chloropropane.

[Generated by AI]

Which product is predominantly formed in this reaction, and why? Are both products formed in equal amounts or is one product favoured over the other? What factors may influence this and how does the presence of substituents on the propene affect the production of a major product?

The major product is 2-chloropropane as it is produced through a more stable secondary carbocation intermediate. As the 2-chloropropane is favoured in this reaction, the two isomeric products are not formed in equal amounts. The formation of major products is influenced by the nature of the carbocation intermediate. The more stable the carbocation intermediate of the reaction, the more stable the

product. The presence of substituents may alter the steric hindrance of propene. Therefore, it can create the reactivity of propene towards HCl and shift the selectivity towards the other product.

The main product of this reaction will more probably be 2-chloropropane because it is formed from a secondary carbocation that is more stable compared to the primary carbocation that is formed to produce 1-chloropropane. The accurate mechanism of the addition reaction of propene and hydrogen chloride is shown in **Figure 4**.

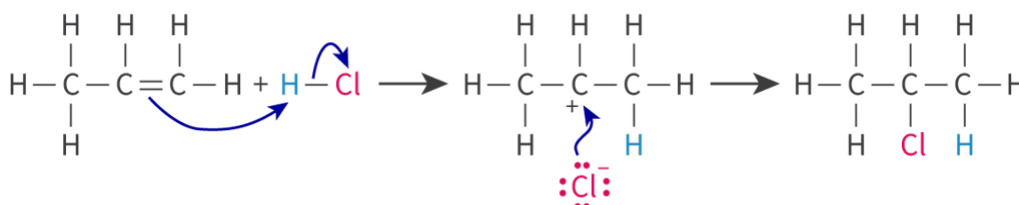


Figure 4. The mechanism of the addition reaction of hydrogen chloride to propene.

 More information for figure 4

The diagram illustrates the mechanism of the addition reaction of hydrogen chloride to propene. It shows the step-by-step formation of 2-chloropropane. The reaction starts on the left with a propene molecule (C₃H₆) depicted with its double bond between the first and second carbon atoms. The hydrogen chloride (HCl) molecule is shown next to the propene with an arrow indicating the movement of the hydrogen (H⁺) towards the double bond. This results in the formation of a carbocation on the secondary carbon as the chlorine ion (Cl⁻) is released. The chlorine ion then bonds with the carbocation forming 2-chloropropane, depicted on the right. The arrows illustrate the flow of electrons during the reaction phases.

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The product from the reaction of an unsymmetrical alkene and hydrogen halide can be predicted using Markovnikov's rule in which it is stated that a hydrogen atom will probably attach to the carbon atom that has more hydrogen atoms already attached, or 'the rich get richer'. The addition of hydrogen chloride to propene allows hydrogen to attach to carbon number 1 (**Figure 5**) as it has two hydrogen atoms attached compared to carbon number 2, which has no hydrogen atoms attached.

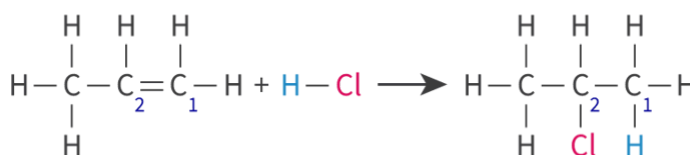


Figure 5. Predicting the major product of an unsymmetrical alkene reaction.

 More information for figure 5

The diagram illustrates a chemical reaction between propene, a type of unsymmetrical alkene, and hydrogen chloride (HCl), demonstrating Markovnikov's rule. On the left side of the equation, propene's structure shows three carbon atoms, with carbon 1 (C1) having two hydrogen atoms attached and carbon 2 (C2) having none. A hydrogen chloride molecule is added to this structure, where the hydrogen atom (H) and chlorine atom (Cl) are marked distinctly.

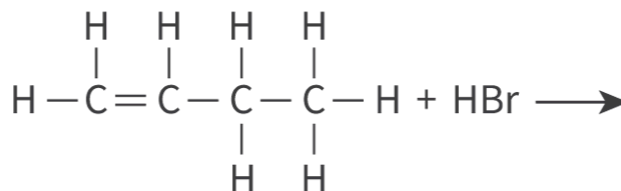
The reaction proceeds with an arrow pointing to the right, indicating the formation of the reaction products. The structure on the right shows the resulting molecule: the hydrogen atom attaches to carbon 1, which aligns with the rule that hydrogen attaches to the carbon atom with more hydrogens ('the rich get richer'). The chlorine atom attaches to carbon 2, completing the reaction product.

This diagram visually explains how Markovnikov's rule predicts the major product in the addition of HX to unsymmetrical alkenes, exemplified by the addition of HCl to propene resulting in 2-chloropropane as the product.

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Worked example 1

Predict and draw the chemical structure for the major and minor products from the reaction shown.

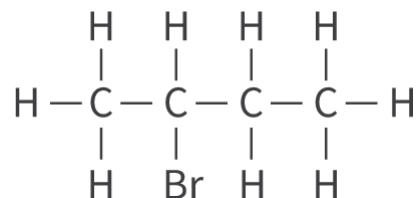


 More information

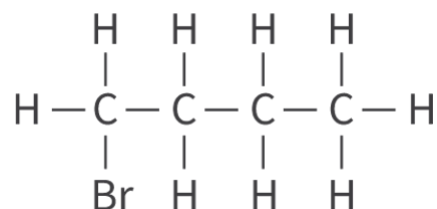
The image displays a chemical reaction with structural formulas. The left side shows the organic compound 2-methyl-2-butene, which has a double bond between two central carbon atoms, surrounded by hydrogen atoms and a methyl group. The plus sign indicates it reacts with HBr (hydrogen bromide), and an arrow points right, symbolizing a chemical reaction leading to the formation of a product.

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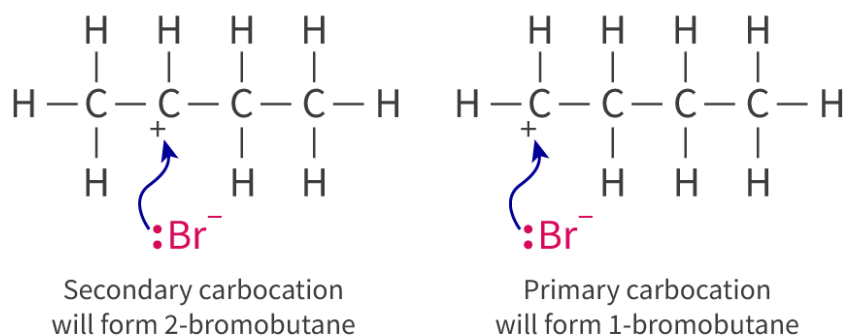
Major product



Minor product



The major product of the reaction between but-1-ene and hydrogen bromide has the Br atom attached to the second carbon in which the minor product has the Br atom attached to the first carbon atom. The secondary carbocation is formed when the hydrogen ion in HBr is attracted to the carbon atom of the double bond, and is more stable compared to the primary carbocation.





Exercise 1

Click a question to answer



1. True or false?



2-iodo-2-methylbutane is favoured in the reaction between hydrogen iodide and 2-methylbut-1-ene.

Check answers

Higher level (HL)

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:**
 - Thinking skills — Designing procedures and models
 - Research skills — Comparing, contrasting and validating information
- **Time required to complete activity:** 45 minutes + 20 minute extension
- **Activity type:** Group activity

Materials

- Modelling clay to represent atom (at least six different colours to represent the atoms of C, H, O, Br and Cl: black for carbon, white for hydrogen, orange for bromine, green for chlorine, red for oxygen, an extra colour for electrons).
- Craft wire to connect modelling clay balls to form the chemical structure and curly arrows.
- Multiple smaller balls of modelling clay to represent electrons — choose a different colour to those used for atoms. They will be attached to the atoms using the craft wire.

Your task

Construct a three-dimensional model that visually represents diverse mechanisms of electrophilic addition reactions such as halogenation, hydrogenation and hydrohalogenation for both symmetric and unsymmetric alkenes. Explain these mechanisms and the movement of electrons and curly arrows using the materials provided. By using online software or a video camera, capture a series of images step by step and create a stop motion animation/film. Then compare these reactions, and present a selection reaction mechanism and its corresponding conditions for symmetric and unsymmetric alkenes to your classmates.

Extension task

Extend your explanation by conducting further research on the application of the reactions. Connect your selected reactions to the nature of science by exploring their coverage, testing, verification and interconnections with other scientific concepts and disciplines.